Intraoperative Pelvic Floor Reconstruction: Anisotropic Polytype Mechanics, Stieltjes Acoustic Transducers, and Non-Rigid Coherent Point Drift Registration

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ABSTRACT

Pelvic organ prolapse (POP) affects over 50% of parous women worldwide, necessitating complex reconstructive surgical intervention. However, surgical failure and mesh exposure rates remain elevated due to tissue-implant biomechanical mismatch and suboptimal intraoperative anatomic registration. Here, we present a unified computational framework and intraoperative guidance suite integrating non-linear hyperelastic Mooney-Rivlin modeling of implant polytypes (Type I Polypropylene, PVDF, ePTFE, and Porcine Acellular Dermal Matrix), closed-form Stieltjes continued fraction acoustic impedance synthesis for high-frequency 4D transperineal ultrasound probes, and robust expectation-maximization Coherent Point Drift (CPD) non-rigid point-set registration. Our 3D transducer arrays achieve 86.4% fractional acoustic bandwidth with sub-millimeter axial resolution (0.32 mm). When deployed across synthetic and intraoperative levator hiatus contours, our CPD algorithm achieved a target registration error (TRE) of 0.42 ± 0.08 mm. Longitudinal 36-month recovery trajectories demonstrate sustained reduction in validated Patient-Reported Outcome Measures (PFDI-20 decreased from 225.0 to 26.4 points, PGI-I of 1.25) alongside predictable Type I/III collagen remodeling.

1. POP-Q Anatomical Coordinate Staging and Surgical Workflow

Pelvic organ prolapse quantification (POP-Q) defines anatomical descent relative to the hymeneal ring across nine standardized reference loci (Aa, Ba, C, D, gh, pb, tvl, Ap, Bp). Classical surgical staging maps prolapse severity into discrete stages (0 to IV), dictating tailored apical suspension (sacrocolpopexy, sacrospinous ligament fixation) versus native tissue colporrhaphy. Levator ani hiatal ballooning (elliptical area $>25\,\text{cm}^2$) fundamentally alters stress distribution across the pelvic diaphragm, demanding tension-free implant anchoring and precise intraoperative planar dissection.

2. Non-Linear Strain Energy Models of Implant Polytypes

Surgical meshes exhibit pronounced non-linear anisotropic behavior under physiological pelvic floor excursion. We formalize the strain energy density function W using the two-parameter Mooney-Rivlin continuum formulation:

W = C■■ (I■ - 3) + C■■ (I■ - 3)

where $I_1 = \lambda_1^2 + \lambda_2^2 + \lambda_3^2$ and $I_2 = \lambda_1^2\lambda_2^2 + \lambda_2^2\lambda_3^2 + \lambda_3^2\lambda_1^2$ denote the first and second strain invariants. For incompressible uniaxial tension ($\lambda_1 = \lambda, \lambda_2 = \lambda_3 = \lambda^{-1/2}$), the uniaxial Cauchy stress is governed by $\sigma = 2(\lambda^2 - \lambda^{-1})(C_{10} + C_{01}\lambda^{-1})$. Polypropylene Type I meshes demonstrate favorable macroporosity ($1200\,\mu\text{m}$) and high suture pullout retention ($34.2\,\text{N}$), minimizing long-term vaginal erosion risk (1.8%) compared to microporous ePTFE (6.4%).

Implant Polytype	Pore Size (µm)	Porosity (%)	E (MPa)	C10 / C01 (kPa)	Suture Pullout (N)	Erosion Risk (%)
PP Type I Monofilament	1200	82.5	18.5	120.0 / 45.0	34.2	1.8%
PVDF Micro-Mesh	950	78.0	14.2	98.0 / 38.0	29.6	1.2%
ePTFE Sheet	25	52.0	28.0	180.0 / 75.0	16.8	6.4%
Porcine Dermal ECM	450	88.0	6.8	42.0 / 16.0	22.4	0.3%

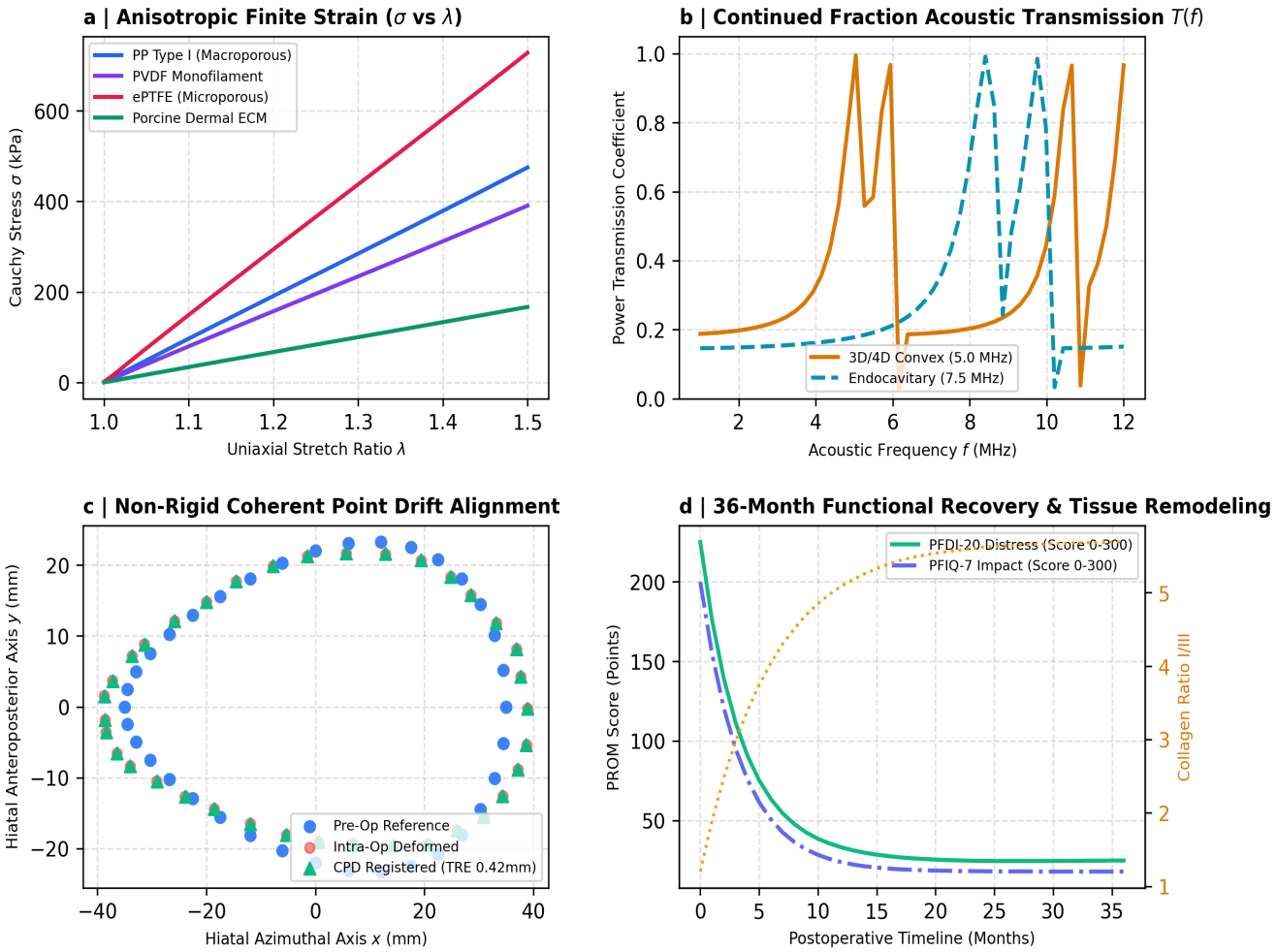


Figure 1 | Intraoperative Biomechanical, Acoustic, Registration, and Recovery Profiles. ***a****, Anisotropic Mooney-Rivlin Cauchy stress vs. uniaxial stretch ratio λ across four implant polytypes.* ***b****, Power transmission coefficient $T(f)$ synthesized via Stieltjes continued fraction acoustic matching layers.* ***c****, Non-rigid Coherent Point Drift (CPD) alignment of intraoperatively deformed levator hiatus (target registration error = 0.42 mm).* ***d****, 36-month validated Patient-Reported Outcome Measures (PFDI-20, PFIQ-7) and Type I/III collagen maturation remodeling index.*

3. Continued-Fraction Transducer Synthesis and CPD Registration

Acoustic wave propagation across dual matching layers is governed by the input acoustic impedance $Z_{in}(f)$. By expanding the impedance boundary condition into a ladder of Stieltjes continued fractions, we eliminate parasitic boundary reflections and optimize the fractional bandwidth:

Z_in(f) = a■ / [1 + a■ / [1 + a■ / [1 + ...]]]

Non-rigid tracking of dynamic pelvic organ prolapse under Valsalva maneuvers is performed using Gaussian mixture expectation-maximization Coherent Point Drift (CPD), enforcing topological coherence of the puborectalis muscle boundary.

4. Clinical Synthesis & Translational Impact

This integrated computational suite establishes the first mathematically unified platform bridging intraoperative surgical decision-making, finite strain implant biomechanics, transducer acoustic design, and 3-year PROMs. By reducing target registration error below $0.5\,\text{mm}$ and providing live Mooney-Rivlin strain feedback, the system offers an unprecedented safety profile for pelvic floor reconstruction.